MOHAN BABU UNIVERSITY

DATA SCIENCE LAB — SAMPLE EXPERIMENT

Experiment: Data Visualization

Student Name: _____

Student ID: _____

Date: _____

Aim

To create and interpret common charts using Matplotlib.

Requirements

Python, Matplotlib, and sample data.

Experiment

1. Create a list of categories and corresponding values.
2. Draw a line chart with title, axis labels, and a legend.
3. Draw a bar chart to compare category values.
4. Draw a histogram for a numeric variable.
5. Draw a scatter plot to show the relationship between two numeric variables.
6. Add grid, labels, and suitable chart titles.
7. Save one chart as a PNG image.

Expected Output

Four labeled charts and one saved PNG file

Conclusion

Write a short interpretation of what each chart communicates

Note: This is a sample practice lab experiment created for learning and website testing. It is not an official university-issued lab manual.